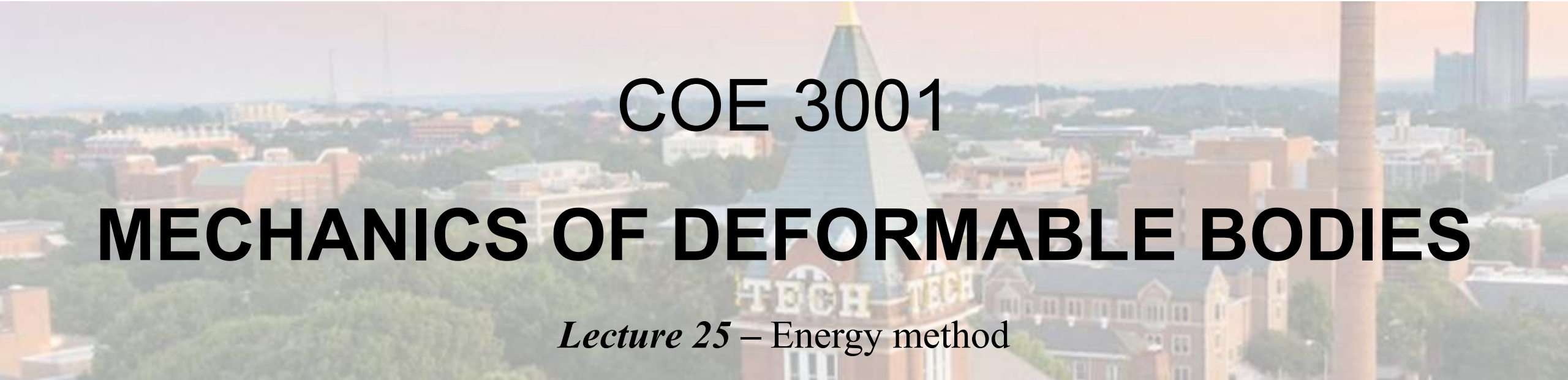




Georgia Tech
College of
Engineering

A wide-angle, slightly hazy photograph of the Georgia Institute of Technology campus. The image shows various red-brick buildings, green trees, and a tall smokestack in the distance. The text is overlaid on this image.

COE 3001

MECHANICS OF DEFORMABLE BODIES

Lecture 25 – Energy method

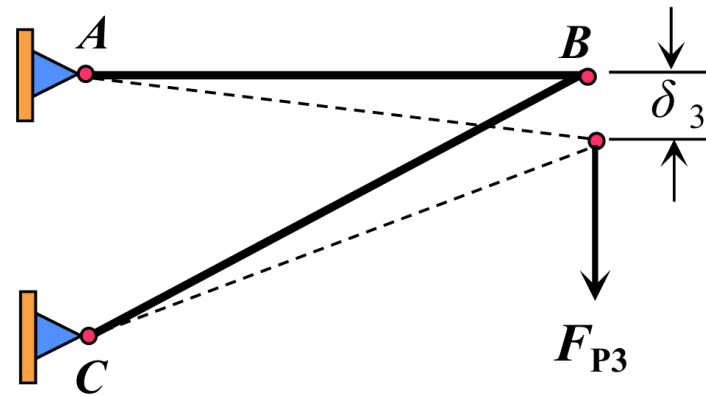
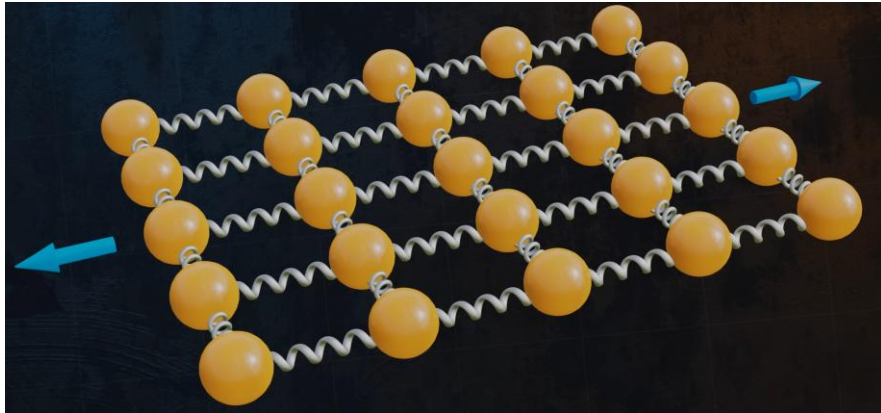
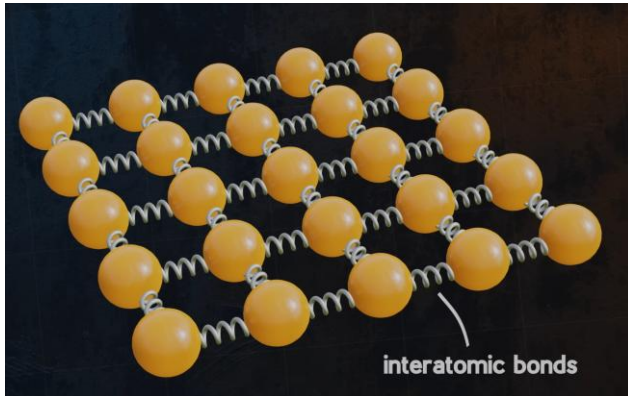
Yazhuo Liu

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Nov. 18, 2025

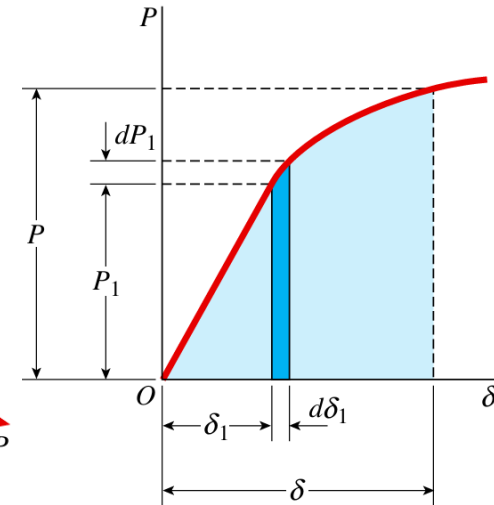
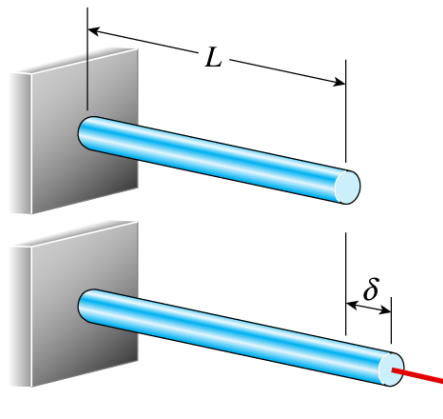
Elastic strain energy

Hard materials store energy primarily as strain energy



Without dissipation and internal source, **external work** fully transforms into elastic strain energy via the **internal work**.

$$W_{\text{ext}} = W_{\text{in}} = U$$

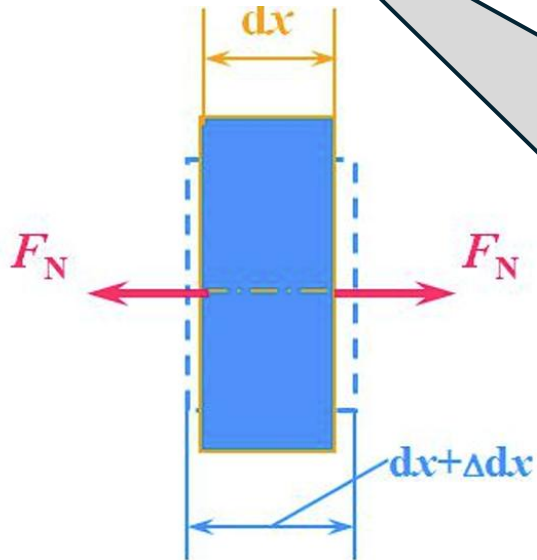


$$W = \int_0^{\delta} P_1 d\delta_1$$

Elastic strain energy

1. Tension/compression

$$dU = \frac{1}{2} F_N \Delta(dx) = \frac{1}{2} F_N \times \frac{F_N dx}{EA} = \frac{1}{2} \frac{F_N^2}{EA} dx$$



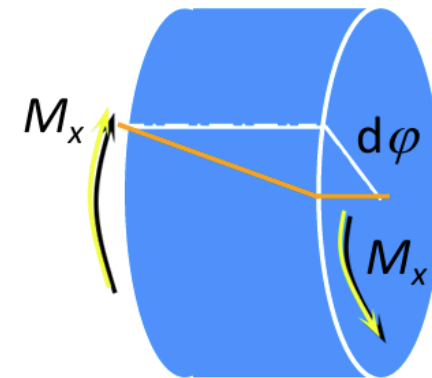
When we consider an infinitesimal element, **we still believe that the force increases linearly with displacement.** Therefore, when the final force reaches F_N and the displacement reaches $\Delta(dx)$, the infinitesimal strain energy is still $\frac{1}{2} F_N \Delta(dx)$

$$U = \int_0^L \frac{1}{2} \frac{F_N^2}{EA} dx = \frac{F_N^2 L}{2EA}$$

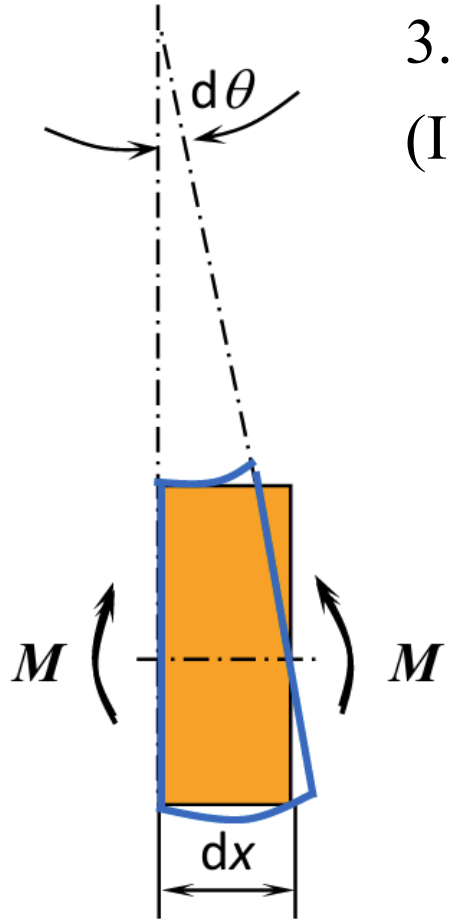
2. Torsion

$$dU = \frac{1}{2} M_x d\phi = \frac{1}{2} M_x \times \frac{M_x dx}{GI_p}$$

$$U = \int_0^L \frac{1}{2} \frac{M_x^2}{GI_p} dx = \frac{M_x^2 L}{2GI_p}$$



Elastic strain energy



3. Bending

(Ignore the effect of shear force)

$$dU = \frac{1}{2} M d\theta$$

$$\theta = \frac{dv}{dx} \Rightarrow d\theta = \frac{d^2v}{dx^2} dx = \frac{M}{EI} dx$$

$$\Rightarrow U = \int_0^L \frac{1}{2} M \frac{M}{EI} dx = \frac{M^2 L}{2EI}$$

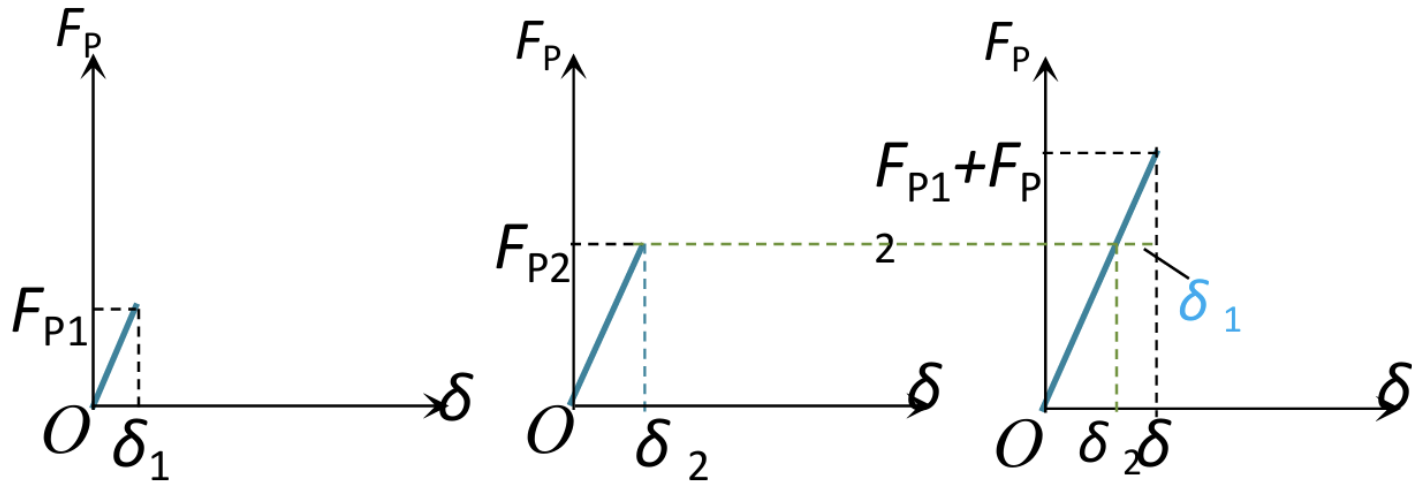
4. Combined loads

$$U = \int_0^L \frac{1}{2} \frac{F_N^2}{EA} dx + \int_0^L \frac{1}{2} \frac{M_x^2}{GI_p} dx + \int_0^L \frac{1}{2} \frac{M^2}{EI} dx$$

or

$$U = \frac{F_N^2 L}{2EA} + \frac{M_x^2 L}{2GI_p} + \frac{M^2 L}{2EI}$$

Limited application for superposition



In linear elasticity region

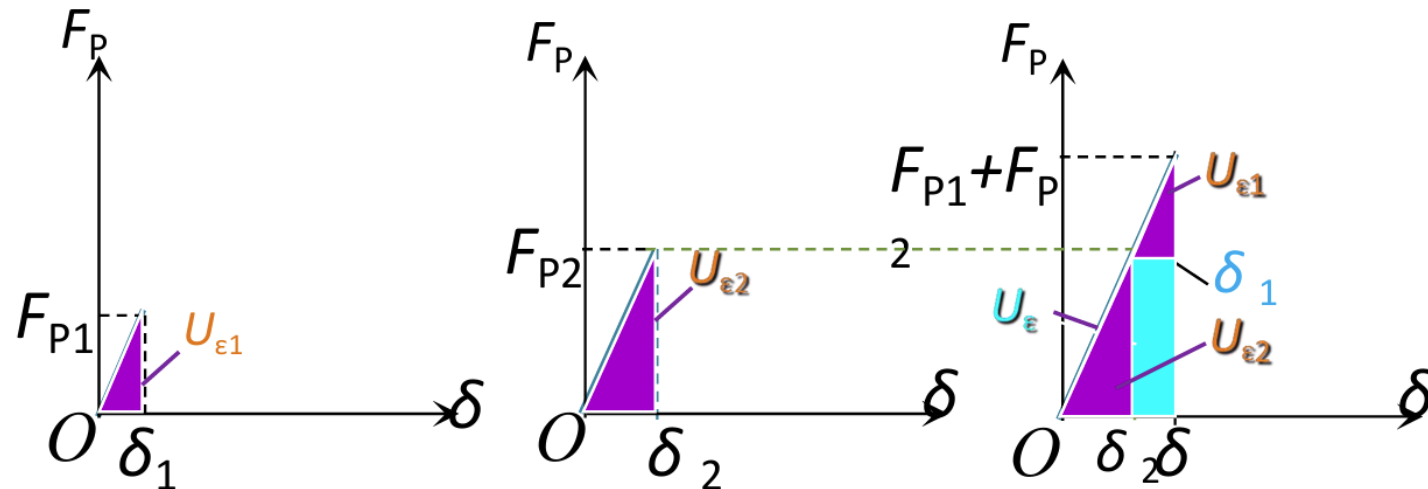
$$\delta = \delta_1 + \delta_2$$

- Displacement can be superposed

But

$$U \neq U_1 + U_2$$

- Strain energy cannot be superposed



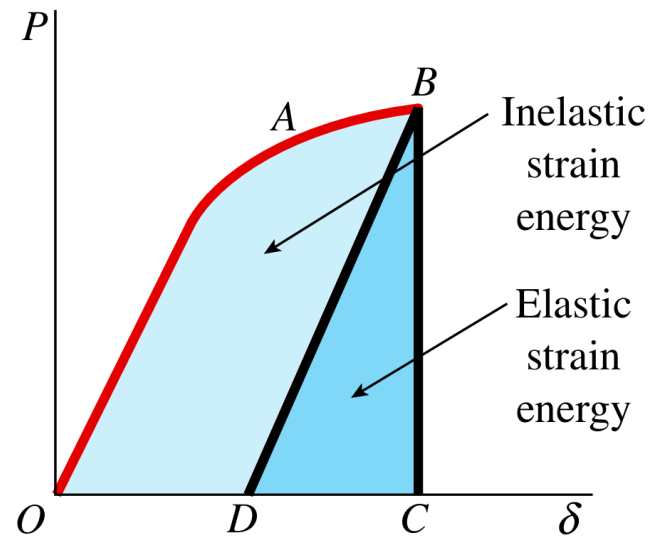
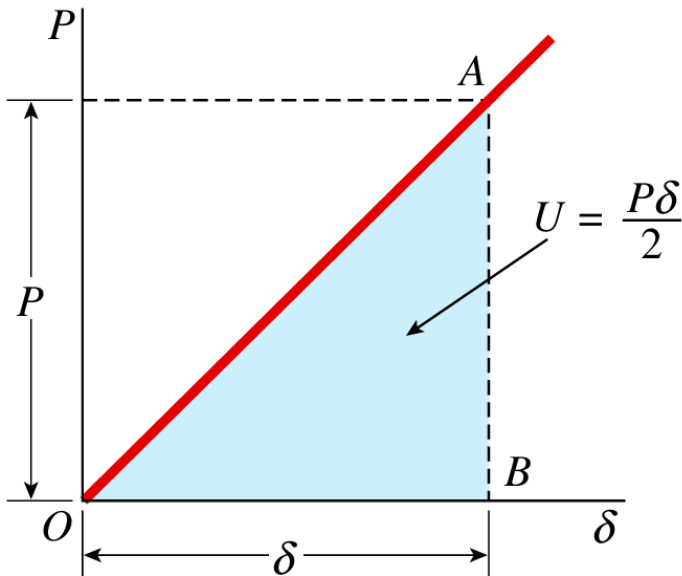
Strain energy density

In the case of a linear elastic deformation

The strain energy density

$$U = \int_0^{\delta} kx \, dx = \frac{1}{2} k \delta^2 = \frac{1}{2} P \delta$$

$$u = \frac{U}{V} = \frac{1}{AL} \int_0^{\delta} P \, dx = \int_0^{\delta} \frac{P}{A} \frac{dx}{L} = \int_0^{\varepsilon_1} \sigma \, d\varepsilon$$



- The total strain energy density resulting from the deformation is equal to the area under the curve to ε_1
- As the material is unloaded, the stress returns to zero but there is a permanent deformation. Only the strain energy represented by the triangular area is recovered.

Example

Determine the line deflection of point C. The rigidity of beam AB is EI .

Solution: AC segment

$$M(x) = \frac{F}{2} x \quad ; \quad (0 \leq x \leq \frac{l}{2})$$

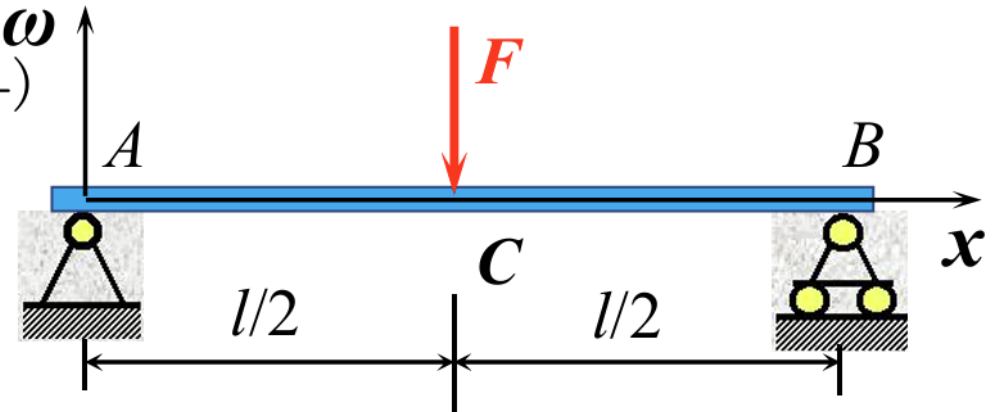
$$U_\varepsilon = \int_L \frac{M^2(x)}{2EI} dx$$

$$U_\varepsilon = 2 \int_0^{\frac{l}{2}} \frac{1}{2EI} \left(\frac{F}{2} x \right)^2 dx = \frac{F^2 l^3}{96 EI}$$

From work-energy principle

$$U_\varepsilon = W = \frac{1}{2} F \omega_c$$

$$\text{So: } \omega_c = \frac{Fl^3}{48 EI}$$



Castigliano's 2nd theorem

ca·sti·lià·no (Italian)

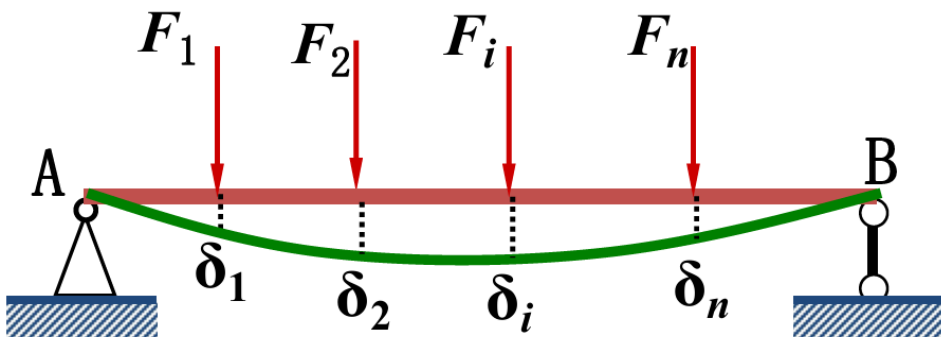
Consider a linearly elastic deformable body or system acting upon by n generalized forces $F_1, F_2, \dots, F_n$. Among all possible equilibrium configuration of the system, the actual configuration is the one for which

$$\delta_i = \frac{\partial U}{\partial F_i}, \quad i = 1, 2, \dots, n$$

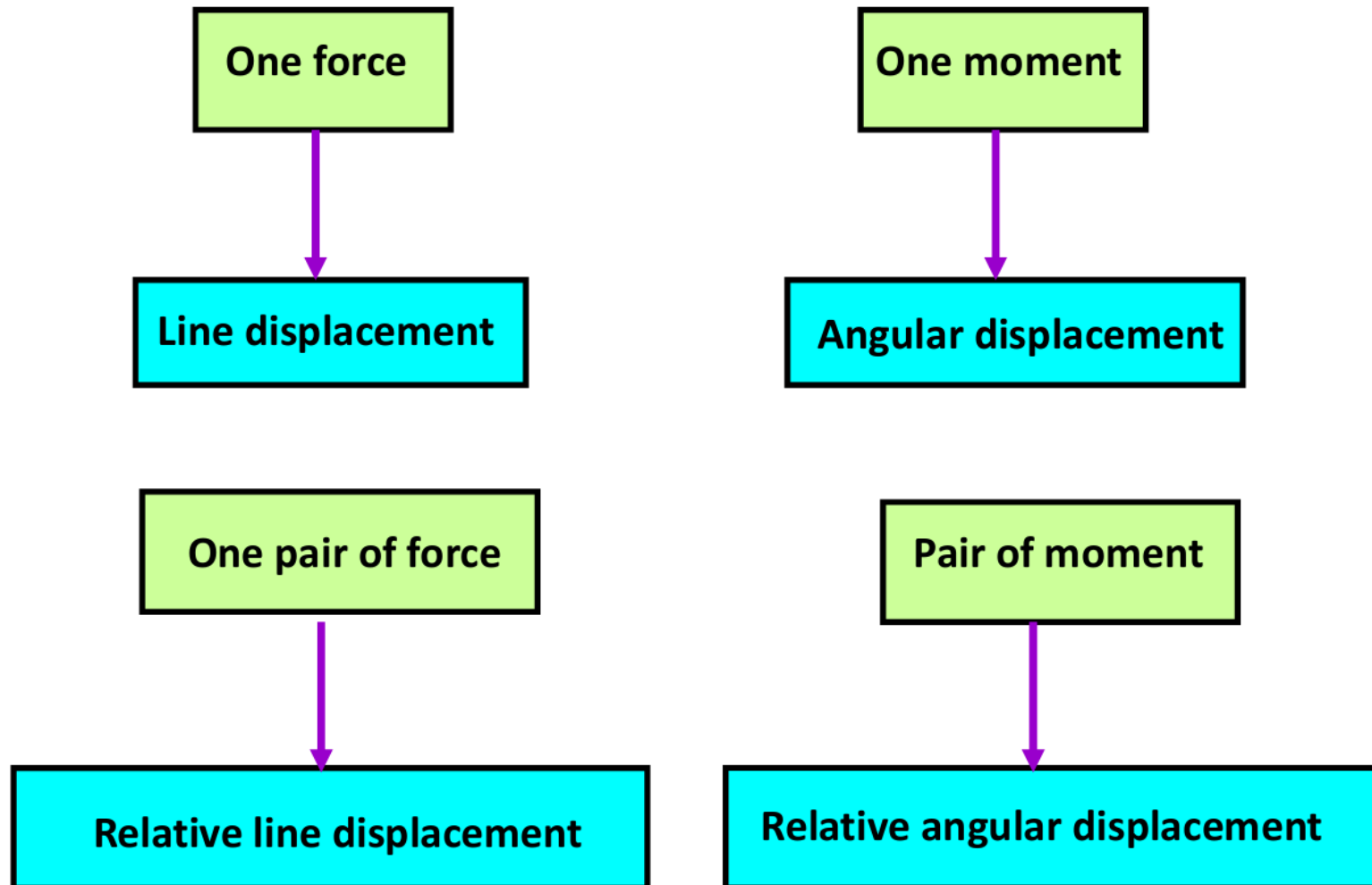
where δ_i is the generalized displacement corresponding to and in the direction of F_i , and U is the strain energy of the system.

If there are n_R redundant generalized forces $R_1, R_2, \dots, R_{n_R}$, then for these redundant

$$\frac{\partial U}{\partial R_i} = 0, \quad i = 1, 2, \dots, n_R$$



Generalized force and displacement



Application of Castigliano's 2nd theorem

Tensile/
compressive

$$\delta_i = \frac{\partial U_\varepsilon}{\partial F_i} = \frac{\partial}{\partial F_i} \left(\int_1 \frac{F_N^2(x)}{2EA} dx \right) = \int_1 \frac{F_N(x)}{EA} \frac{\partial F_N(x)}{\partial F_i} dx$$

Torsion

$$\delta_i = \frac{\partial U_\varepsilon}{\partial F_i} = \frac{\partial}{\partial F_i} \left(\int_1 \frac{T^2(x)}{2GI_p} dx \right) = \int_1 \frac{T(x)}{GI_p} \frac{\partial T(x)}{\partial F_i} dx$$

Bending

$$\delta_i = \frac{\partial U_\varepsilon}{\partial F_i} = \frac{\partial}{\partial F_i} \left(\int_1 \frac{M^2(x)}{2EI} dx \right) = \int_1 \frac{M(x)}{EI} \frac{\partial M(x)}{\partial F_i} dx$$

Combined
load

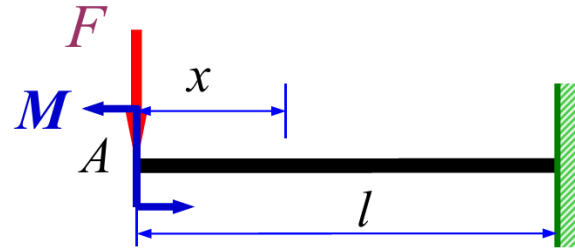
$$\begin{aligned} \delta_i &= \frac{\partial U_\varepsilon}{\partial F_i} = \frac{\partial}{\partial F_i} \left(\int_1 \frac{F_N^2(x)}{2EA} dx + \int_1 \frac{T^2(x)}{2GI_p} dx + \int_1 \frac{M^2(x)}{2EI} dx \right) \\ &= \int_1 \frac{F_N(x)}{EA} \frac{\partial F_N(x)}{\partial F_i} dx + \int_1 \frac{T(x)}{GI_p} \frac{\partial T(x)}{\partial F_i} dx + \int_1 \frac{M(x)}{EI} \frac{\partial M(x)}{\partial F_i} dx \end{aligned}$$

Example

Determine the line and angular displacement of point A by Castigliano's 2nd theorem.

Solution: 1) determine the ω_A

$$M(x) = -Fx, \quad \frac{\partial M(x)}{\partial F} = -x$$



Based on Castigliano's 2nd theorem

$$\omega_A = \frac{\partial U_\varepsilon}{\partial F} = \int_l \frac{M(x)}{EI} \frac{\partial M(x)}{\partial F} dx = \int_0^l \frac{(-Fx)}{EI} \times (-x) dx = \frac{Fl^3}{3EI}$$

$$M(x) = -Fx - M, \quad \frac{\partial M(x)}{\partial M} = -1$$

$$\theta_A = \frac{\partial U_\varepsilon}{\partial M} = \int_l \frac{M(x)}{EI} \frac{\partial M(x)}{\partial M} dx = \int_0^l \frac{(-Fx - M)}{EI} \times (-1) dx$$

2) determine the θ_A (Additional moment M)

$$M(x) = -Fx - M, \quad \frac{\partial M(x)}{\partial M} = -1$$

Set $M = 0$, so

$$\theta_A = \frac{Fl^2}{2EI}$$

Example

Determine the displacement of point A. EI is the rigidity.

Solution: set $F_A = 2F$, $F_B = F$

$$AB \text{ seg: } M(x) = -F_A x, \quad \frac{\partial M(x)}{\partial F_A} = -x$$

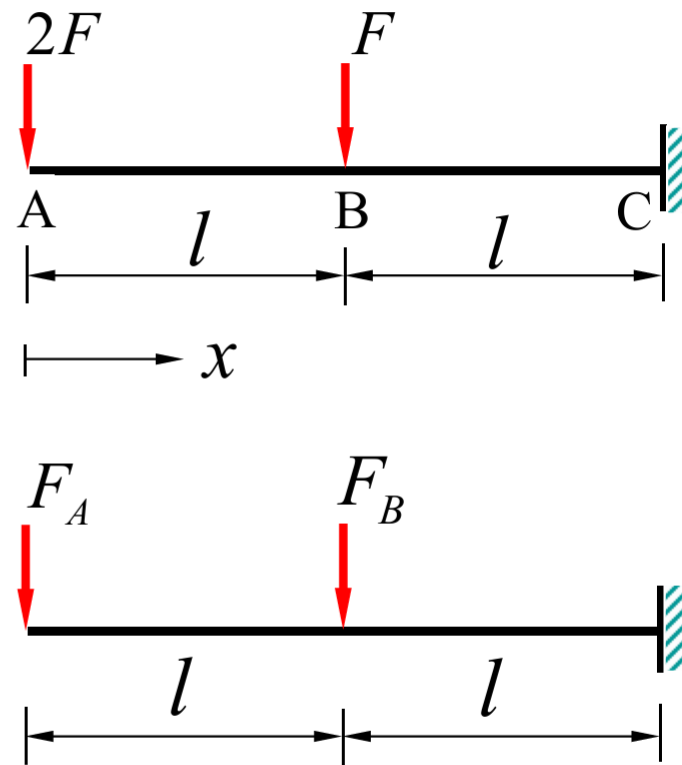
$$(0 \leq x \leq l)$$

$$BC \text{ seg: } M(x) = -F_A x - F_B (x - l),$$

$$\frac{\partial M(x)}{\partial F_A} = -x \quad (l \leq x \leq 2l)$$

$$\omega_A = \frac{\partial U_\varepsilon}{\partial F_A} = \int_l \frac{M(x)}{EI} \frac{\partial M(x)}{\partial F_A} dx$$

$$= \int_0^l \frac{(-F_A x)}{EI} \times (-x) dx + \int_l^{2l} \frac{[-F_A x - F_B (x - l)]}{EI} \times (-x) dx = \frac{37Fl^3}{6EI}$$



Example

EA is known, determine the horizontal displacement of point A and the rotation angle of AB.

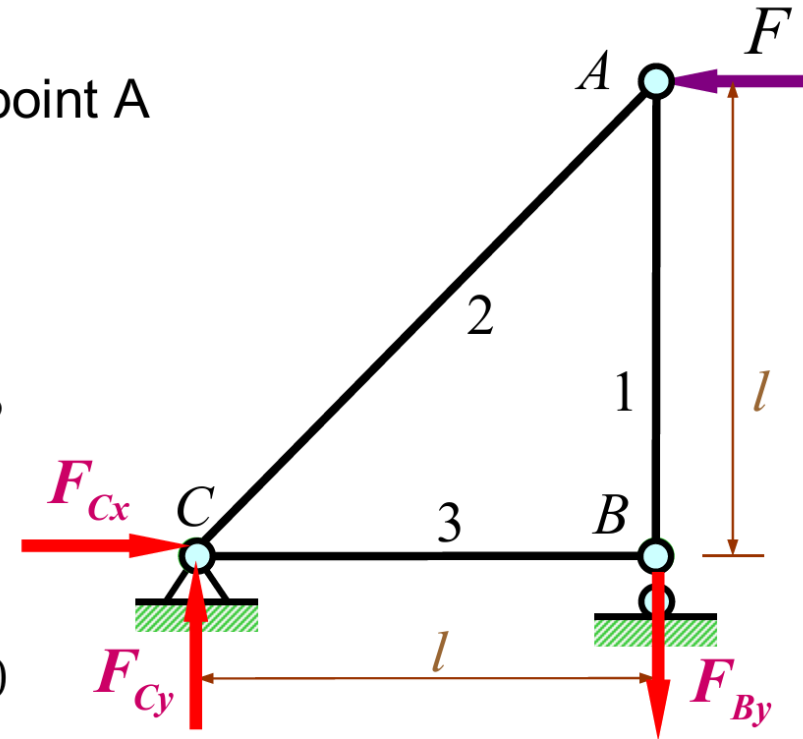
Solution: 1) horizontal displacement of point A
From force balance,

$$F_{By} = F_{cy} = F_{Cx} = F$$

From the equilibrium of point A and B

$$F_{N1} = F, \quad F_{N2} = -\sqrt{2} F, \quad F_{N3} = 0$$

$$\frac{\partial F_{N1}}{\partial F} = 1, \quad \frac{\partial F_{N2}}{\partial F} = -\sqrt{2}, \quad \frac{\partial F_{N3}}{\partial F} = 0$$



$$\delta_{Ax} = \frac{\partial U_{\epsilon}}{\partial F} = \sum \frac{F_{Ni}}{EA} \frac{\partial F_{Ni}}{\partial F} l_i = \frac{1}{EA} \left[Fl + (-\sqrt{2}F) \times (-\sqrt{2}) \times \sqrt{2}l \right]$$

$$= \frac{(1 + 2\sqrt{2})Fl}{EA} (\leftarrow)$$